INCH-POUND
MIL-DTL-24643/62A
28 January 2010
SUPERSEDING
MIL-DTL-24643/62
1 March 2007

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LSPB2SD
(INCLUDING VARIATIONS LSPB2SDW, LSPB2SDOS, LSPB2SDOSW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable is a single or double shield twisted pair intended for use with Profibus data communication systems. This cable can be manufactured in four variations: LSPB2SD (single shield with drain, non-watertight), LSPB2SDW (single shield with drain, watertight), LSPB2SDOS (double shield with drain, non-watertight), LSPB2SDOSW (double shield with drain, watertight).

Construction (Waterblocked and Non-Waterblocked)

First	-	Copper conductor, solid, 22 AWG nominal.
Second	-	Polyolefin insulation. Wall thickness as necessary, in order for the completed cable to comply with the specified electrical requirements.
Third	-	Two insulated conductors, twisted together with appropriate fillers to make a round cable such that they comply with the electrical requirements specified herein. Identification codes for pair − red and green.
Fourth	-	For watertight variants, waterblocking material shall comply with the requirements of 3.3.5 and 3.3.5.3 of MIL-DTL-24643.
Fifth	-	Optional binder.
Sixth	-	24 AWG nominal tin-coated copper drain wire.
Seventh	-	Round, tin-coated copper braid, optimized for 01 and 02 variants.
Eighth	-	Optional binder or optional waterblocker.
Ninth	-	Cross-linked polyolefin jacket, minimum thickness of 0.040 inch for the 01 and 02 variants. Overall diameter (OD): 0.360±0.030 inch for the 01 variant, 0.375±0.030 inch for the 02 variant (this completes the 01 and 02 variants).

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Construction Variant for Types LSPB2SDOS and LSPB2SDOSW

Tenth	-	For 04 variant: Waterblocking material in accordance with 3.3.5 and 3.3.5.3 of MIL-DTL-24643.
Eleventh	-	Optional binder or optional waterblocker.
Twelfth	-	Round, tin-coated copper braid, optimized.
Thirteenth	-	Optional binder or optional waterblocker.
Fourteenth	-	Cross-linked polyolefin jacket, minimum wall thickness of 0.040 inch (this completes construction of the 03 and 04 variants).

TABLE I. Details.

Military part no. M24643/62	**Type**	**Number of pairs**	**Overall diameter (inches)**	
			Min.	**Max.**
-01	LSPB2SD-1	1	0.330	0.390
-02	LSPB2SDW-1	1	0.345	0.405
-03	LSPB2SDOS-1	1	0.440	0.500
-04	LSPB2SDOSW-1	1	0.450	0.530

REQUIREMENTS:

Qualification Required.

INSPECTION:

Basic Electricals:

DC resistance (ohms/100 m, max.)	5.45
DC resistance unbalance (percent, max.)	5
Jacket flaws	No failure
Conductor and shield continuity	No failure
Voltage withstand	
Conductor to conductor	1.5kVAC or 2.5kVDC
Conductor to shield	1.5kVAC or 2.5kVDC

Group A:

Visual and dimensional	No failure
Watertightness (Types LSPB2SDW-1 & LSPB2SDOSW-1 only)	No failure
Capacitance	
Mutual capacitance of a pair (max. @1 kHz)	3.0 nf/100 m
Characteristic impedance (0.2 MHz to 20 MHz)	150±15 ohms

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TABLE II. Swept electrical parameters.

Frequency (MHz)	0.2	4.0	16	31.25
Attenuation dB/100m (max.)	1.0	2.5	5.0	7.2

Group B:	
Drip (95±1 °C)	Zero
Tear (lb/in thickness, min.) (ASTM D470)	35
Physicals (unaged)	
Insulation	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	160
Jacket	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Thermoset proof test (percent, max.)	
Jacket	50
Group C:	
Physicals (aged air oven)	
Insulation (polyolefin)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (136 °C for 168 hours)	
Tensile strength (percent retained, min.)	60
Elongation (percent retained, min.)	60
Shrinkage	No failure
Cable filler removability	No failure
Heat distortion (percent of unaged, max.)	30
Conformance to material, construction and coverage	No failure
Surface transfer impedance	
Milliohms per meter (max.)	700
EMP response (dB, min.)	60
Permanence of printing (jacket) (cycles, min.)	125

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Group D:

Flame propagation (cable)	No failure

Qualification Inspection:

Qualification inspection shall include basic electricals; groups A, B, C and D; plus the following:

Aging and compatibility (cable) (95±2 °C)	No failure
Abrasion resistance (jacket)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	8
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index, max.	
Jacket	25
Fillers	55
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – Type A (hardness, min.)	80
Watertightness (max. leakage, cm^3)	65.5
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSPB2SD, LSPB2SDW, LSPB2SDOS, LSPB2SDOSW	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2009-105)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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